Unit 9 - Computer Networks
Assignment 13: Managing a LAN Task 5

Introduction

Task description - *You are a junior consultant for T5 Alliance. The work you have done so far for Competition Bikes Ltd has been very well received. As they have no IT support staff of their own, they are interested in outsourcing the ongoing management to your company. First, however, you must prove your knowledge of network management.*

Task 5

(U9P5) - Outline the purpose of networking tools

Task description - *Networks are used to provide services to end users. Describe the following services.*

- *Directory Services (for example, Microsoft Active Directory)*
- *Email (for example Microsoft Exchange, Lotus Notes)*
- *Remote Access*
- *File sharing (for example, FTP, BitTorrent, OneDrive/Dropbox)*
- *Applications Services (for example, Databases, web servers, proxy servers)*
- *Shared Resources (for example, printers)*

Introduction

For this document, I will be outlining the purpose of networking tools within a network. I will carry this out by mentioning and explaining the purpose of network services, while some may be more detailed than others due to their sophistication/simplicity. Also, I will be mentioning how they are able to carry them out and a few advantages and disadvantages the services bring.

To define a network service, it is essentially a type of application which is found on the network layer on the OSI model and above. Network services are often used in a Client/Server Network or a Peer to Peer network, where the network services have the purpose of providing storage and manipulation of data shared in the network. While also providing presentation and communication for the users within the network.

Usually, network service tools are controlled by a central server. Meaning that, if a user has sent an email to other employee, the email would go to the central server before its sent to recipient. Therefore, it is essential that network services allow different tasks, such as: access of the files that are shared on the network or even a user logging onto the network.

Basically, network services are the applications that would provide services to different users on a computer network. Giving the network the ability to work efficiently as an infrastructure, offering value to the business process of keeping data integrity and a constant workforce over the said network.

Directory Services

A directory service is a software system which stores, organizes and provide access to the information from a computer operating system. Essentially, it is a type of network service which would help make all the resources on the computer network available to the user, as well as different applications.

Usually, this type of service would store all the information about the computer, such as usernames and passwords, in a database. In order to access the information, the Lightweight Directory Access Protocol (LDAP), is a protocol is used to enable any user to locate data such as: organisations, individuals and other resources such as files and other devices in a network.

This is essential so the information can be stored in a secure location and accessed with ease, due to the arrangement and usability of data, bringing multiple benefits for users on a network. Since it is an easy to use organised data service, this service could also help to manage the security of data in a network.

On a larger network, when multiple computers are needed to be identified, administrators use domain control such as: usernames, passwords and authentication management. Meaning that servers need to be authorised by a domain in order to manage larger networks.

An example of a directory service application is Microsoft Active Directory. Which is used to carry out automated network management of user data, security and other distributed resources. Also, it has the ability to carry out internal operations with other directories.

Essentially, it is designed to distribute networking environments. Some examples of this include email addresses, computer communication and other resources connected to the network which can be accessed by users.

In order to access the stored information found in directory applications, such as the Microsoft Active Directory, requires some form of service in order to translate, identity and keep the data secure. Examples of directory services include: DNS, authentication management and account management.

Domain Name Server, or DNS, is the service that helps translate the IP address into domain names. The IP addresses are written in numerical figures separated by a decimals, so it can be used an identification of seperate machines, whether in the same network or different networks.

While the domain names are written in alphabetical characters, this is so they would be easier to remember for a user, instead of only the machine keeping secure storage of the name.

Essentially, the DNS would help store the domain names in a large database and map them to each IP address. If an IP address is not found, the DNS would keep requesting for the intended address until the correct IP address is located.

Meaning that having a domain name server is vital, as it would enable users to access web pages, as it would look for the IP address for the domain name for the website that the users are trying to access.

If there was no DNS, then it would mean that there would be no easy method of accessing Internet, meaning that it would be unreliable and most likely be shut down. Some domains that users may have request for includes: .com, .org, .gov and countless amounts of other combinations. Basically, the DNS exists to make sure that if users have multiple requests for the same domain name, which it would then help the access of data in a network.

Authentication management is basically the identification of who the user is when they are logging on to the network system. Therefore meaning that the use of use of usernames and passwords is essential for identifying individual users.

A benefit of authentication is that there would be an increase in accuracy and security on the network. The reason is simply due to the fact that users are recommended to have their passwords written without the knowledge of other users to said password, while also making sure that they can remember it.

However this isn't a very strong method of security, making it potentially dangerous for certain companies, especially with companies that have transactional systems, such as bank accounts. Meaning there could be potential risks in authentication management.

For instance, a risk could be that the password could get stolen, either due to poor encryption or poor secrecy of the password. Meaning that all the user details and information could get stolen as well, with the risk of other data within the network depending on how much the user can access.

If this was the case, the authentication management system would help to ensure that the passwords are updated. This is so users could ensure that their passwords are updated constantly for security reasons, if another anonymous user has already figured out the password or if the password isn't strong enough.

Account management is the basic responsibility of keeping control of the different user accounts within a network. For instance, as when a user wants to change their username, the admin in control of management can either enable or disable the access to do so in a simple manner.

Another instance where this would be used is the management of a single user or multiple users ability to access the computer and the network.

In the directory service, the account management would store all the network accounts in a long list. Then, when all the accounts are stored, a server would be able to help by configuring the service passwords, then updating the changes to the stored account. Also, an account management would also keep a list of all the files that a specific user has stored.

Overall, directory services are network services which helps store, organise, identify and secure resources such as personal data, company data, email address and the ability of providing access to users.

Basically, this service is a network service,due to the fact it can identify network resources (i.e. email address). So the computers and other peripheral devices on the network can obtain the network resources, making them accessible to the user.

Email

Emails is simply known as electronic mail, this service allows the user to messages to other people through the network. To elaborate, emails allow users to write electronic messages and enable them to send them through the mail server. Which would then organise and identify the email data, making sure that the email goes to the right computer.

With email, users are able to send formatted text and images to other users. Email is still common method of telecommunications that is used widely around the world. However, considering that email is mostly used for text and images, the user also has the ability to attach videos, files and other documents.

An example of an email service would be Microsoft Exchange and Lotus Notes. These services are simple in use due to the simple nature of emails, so there isn't a significant difference in how the two services function, only with the GUI being the major difference.

Essentially, both services carry out the simple purpose of an email service (which is to send emails and open received emails), but have different graphical user interfaces due to the software designs being different.

However, the use of email services may be similar, but there are many emailing protocols that are associated with email. With protocols such as: Simple Mail Transfer Protocol (SMTP), Post Office Protocol (POP) and Internet Message Access Protocol (IMAP).

The purpose of a SMTP server is that it ensures the email is being sent to the correct location, this is so a significant amount of email packets wouldn't be lost within the transfer process.

This is carried out once the user has sent the email, which the email client would then connect to this server.

Where it would inform the server the address of the user and that an email is being sent over the network. After this process, the email would then need to be connected to another domain in order for it to be successfully sent in. Which it would then contact the user's Domain Name Server, or DNS.

Therefore allowing the user to have the IP address for the email server of the user that it is being sent to. The reason for this is because it enables the user's SMTP server to connect to the user (that is being sent the email) to their SMTP server, which would then allow the email to be finally sent to the user.

However, a disadvantage for the SMTP is that it is known for users receiving significant spam emails, since it lacks the ability to verify who sent the email. Meaning that there is vulnerability to unauthorized users sending malware such as viruses through email attachments.

IMAP allows users to access their email messages on a server without the need to download them to their hard drive. Both IMAP and POP3 are similar in functionality. With the main difference between the two being users need to download messages to their hard drive before actually reading them whilst using POP3.

However, using an IMAP mail server would prove to be more beneficial for users, as they would be able to view all their email messages on multiple workstations and always see the opened emails. For example, if a user deletes an email from their computer then they would still be able to view this email on their phone. The reason for this is because the emails don't get automatically deleted from the server.

However, the drawback of IMAP is that if users don't have access to the Internet, then they would not be able to view their emails. Which is becoming less of a disadvantage due to the fact there is more access to the internet in the modern day than any time period in history.

Therefore, on the modern day, examples of where email is commonly used would be in offices or in schools, where important information would need to be sent to the recipients. However, emails are used for business uses more than personal uses, due to the fact instant messaging has made email obsolete for personal use.

Although it doesn’t mean emails are obsolete or lack importance in other areas. This is due to the fact email sends data over a network, which is sent to the recipient accounts in seconds. Moreover, they can also use emails on number of devices such as Smartphones, Tablets and Laptops.

Which is a benefit of using email, which is that emails are very easy to send. Therefore meaning that they are significantly faster than a traditional post, also due to the fact that the user can use email 24/7. Therefore, the user can send emails to someone without location or time being a barrier.

Overall, email is a quick and effective way of sending a message through computer networks, rather than using a traditional post. It is one of the most common uses of telecommunications for the past decades, ever since the internet was first available for the public.

This is due to the fact that email is a cheap service when using broadband internet connection, which is the most common form of internet connection from an ISP. Meaning, if users have an account with an Internet Service Provider, they can use the features of an email without being charged for, so users don't lose any money.

While dial up users would need to pay charges at local rates to send emails to someone, it is no longer an issue. Since dial up isn't very common and almost extinct in the modern day, almost voiding this drawback about email use.

However, a still applicable drawback of email use is that the recipient would need to have a internet connection to access and use emails. Which is also another dying disadvantage due to the wide availability of an internet connection.

Remote Access

Remote Access is the ability to access a user's personal computer from a remote location. To elaborate, Remote Access is when an administrator gains access to a computer or a network, without actually being on the device itself, instead obtaining access to said device from a remote distance.

Essentially, Remote Access is another feature of a telecommunications service. It contains the capability to approach a user's computer from a distance. With the help of programs such as Remote Access (found in MacOS) or PCAnywhere (found in Windows OS), permitting consumers to supervise outlying nodes by their native device.

An example of a use for this is within in a business. For instance, if an employee is struggling to operate certain software in a workstation, the administrator could get access to the workstation to fix the issue that occured from a distance, without the need to waste valuable time trying to locate the workstation in the building.

In order to achieve a remote access relation successfully, the native device needs to contain the remote client software constructed in place. While the remote machine requires containing the remote server software constructed in its position as well. Additionally, for users, a password

and username is asked constantly in order to certify the conjoined consumer, usually the administrator.

Usually, there are two types of remote access. The first method is done by sharing a session, which is when users can see what the administrator is doing as they have control over the user's computer/device. The second method is when the computer is locked to the original user, while the administrator, which is using remote access, has the only access to the the computer/device.

With both methods, the remote access software responds very quickly to the network server. However, this is due to the exclusive variance amongst a remote host. Meaning if conjoined workstations are remotely accessed immediately in the network, it would affect the performance of information to a much slower pace.

However, remote access is not all about a user containing the capability of being conjoined to a remote machine, it defines that a user contains the capability of being able to supervise the machine as soon as the link has been completed.

A remote access program can simply convert a user's device into the distant computer the administrator gains access to. This is beneficial for consumers who operate in their own homes and for the server administrators who constantly need to modernise, form alterations on their server machines or to fix any issues that can appear in user's devices.

Additionally, the majority of remote access programs also contain the ability to permit consumers to send files among the native and distant machines, which can spare a significant amount of time that would therefore limit the work output.

In terms of data security, a remote access server commonly contains a firewall server to assure safety and a router which involves passing on the remote access demand to a different segment of the collective network.

The server may also operate with a modem pool manager, therefore to allow a tiny cluster of modems to be divided between an extensive numbers of intermittently current remote access consumers.

Overall, remote access is the simple act of taking over a users device as the administrator, which could be done due to an issue with the software on the user's device. Remote access can be very beneficial and supportive for a network, only when used correctly however.

What I mean by this is that it is important for the administrator to not empower it, unless it is necessary, as it can add another security concern that the consumer will have to allocate with the administrator. For instance, the benefits of remote access for a business point of view is that it would save the employees costs on travelling and office space, another benefit for remote

access in a business is that the information would be collected quicker and makes the business respond to its customers and it would increase their sales of the business.

File sharing

File sharing is a service which allows the user to share their file to different computers, so that other user can access their files and data. To elaborate, File sharing is other a private and public distribution of information and data capacity in a network, with certain files containing different standards of access privilege. Although other users could save the files onto a hard drive and send them to another user.

Essentially, file sharing allows the user to write and read the files from any machine that the software supports. Usually, they would need to have some amount of space to save their personal files and documents. Making file sharing different to file transfer, as a user utilising file transfer will discover that all files just depart by a single node to a different one.

File sharing has been used as a featured mainframe and multi-user computer systems for many years from the current date. Even though File sharing wasn't very popular when it was introduced, it has become much more popular over time. Due to the fact that internet connections are much more frequent and more reliable than it previously has been.

Currently, there are two main procedures of file sharing. The first method is peer to peer file sharing, which is essentially the availability of data being uploaded by a user which would be downloaded by a range of users. Since it's from a user, there are some files that are too large in capacity for the device the user currently has.

Which is why data in peer to peer file sharing is divided into segments, to make uploads and downloads easier. These segments are then rearranged on the consumer's device. However, this makes downloading slower for the user due to the fact the individual segments have to be downloaded, especially if there are more users connected.

The second kind of File sharing is a file server, which defines that the files are hosted onto a single main server, often referred to a cloud. This contains vast amounts of capacity for storage and performance. Meaning that any user has the ability to download the file by the server, without the performance of their machine being affected.

Which is the method of one popular file sharing service, which is the Google Drive service. Essentially, Google Drive allows users to read and write their documents and files, without having to worry about saving, as it would automatically save the user's documents every time a keystroke is detected.

In addition, users would be able to access their work from Google Drive from any device, as it stores the work on their could server. Meaning that it can be accessed on a PC, laptop, smartphone, tablet and other mobile devices that support the protocols.

Google Drive also allows users to share their work with others, allowing other users to edit, comment or read the document. Other examples of file sharing services are DropBox and OneDrive. These file sharing services are similar applications which, like Google Drive, would also store users, files(such as their pictures and documents) and would allow users to access these files from anywhere.

In terms of user interface, they work the same way as Microsoft's file explorer would, but it is stored in a cloud which can be accessed by multiple machines. Meaning that file sharing can also be shared outside of the network, in addition to many other files can readily be portioned outside a network. Instead of being stored in a single hard drive that could only be accessed by one machine at a time.

For the user, file sharing can be done with ease, either by just uploading a selected user file on the cloud or another form of data storage for the file sharing service. Permitting many users to utilise the identical file, or file from a certain application, giving multiple users the ability to read, write, duplicate or print the file. Meaning that a file sharing operation can contain more than one user.

In addition, all users can contain the identical or diverse standards of access privilege. Meaning that file sharing can as well define containing an appropriate quantity of individualised file perservitation in an ordinary file system, which increases the security for sensitive information.

One advantage of file sharing is that they user would be able to backup their files without affecting storage space on their computer. This means that a previous version of the user's computer can be restored if a vital error occurred, without significantly affecting the storage capacity of the user's device.

Another advantage would be that there would reduce the number of programs on any given network. Meaning that there will be a lot more storage space for higher priority applications and files, or if the user's device is at capacity, could use the cloud storage as a secondary hard drive.

The disadvantages of file sharing is that there are security problems, due to the fact that different users can access the file and documents from the internet. This makes the integrity of the data at risk when stored in a cloud accessible by any device, instead of a hard drive stored in a single machine.

Another disadvantage would be data would be redundant, this means that the information and files have the risk of going corrupt due to the fact they have been moved or changed significantly.

Overall, File sharing is the ability to access files, programs, images, video and other programs with other users or on a different device. Essentially, the user can share their data with public with more than one user as well as privately with one or two or fewer users.

Another reason is is used is due to the fact that file sharing makes it easier for user to backup files and reduce the number of programs on any given network. For instance, when a user creates a file, they would decide and select the people that can access this file. This is so a group of people can do a combined work by reading, viewing and modifying the same file at a same time. This could be done by online sharing such as Google drive.

Basically, this is a secure and quicker way for people to access files and share information within the organisation. Hence why file sharing is mainly used in organisations and educational purposes. Companies use file sharing to share their presentation and other important documents.

Applications Services

Application services manages software based services and providing solutions to user over a network. To elaborate, application services are third party services that collects and stores software data, work information and user information over a Wide Area Network by a centered information location, such as a server.

Basically, these services provides communication used in peer to peer network or server network. They also allow the user to access files on the network and other software applications. Application software helps the user to perform different tasks. This could be for business, educational or general purposes.

An example of an application service is a database application. A database service is an application software which allows a manager to keep storage of important data together in one system/form saved in a computer or other forms of storage.

Essentially, the purpose of databases is that a computer program can immediately choose requesting segments of information and many users consider databases as an electronic filing operation.

Database software are used by businesses to keep the record of their sales as well as customer details, which is vital to keep so the business known what to benefit from with the information. This is done by allowing the manager to create forms, queries and reports, which helps keep the database organised and structured so the information can be easily accessed by the manager.

Another example of an application service is a web server. Basically, this is a service that enables us to connect to the internet through an application that is stored on the computer, these applications would include Google Chrome, Mozilla Firefox and Internet Explorer.

A website is usually run by a web server, which is almost always found in a network and includes the connection to the internet. A web server is essential to connect to the internet, hence why there are internet service providers. Hence why a web server can be found in a WAN.

This is due to the fact that the majority of connections to a LAN server is limited. As the internet is used by people around the world, it cannot be provided with LANs. Making it only available through one source, which is by an internet service provider.

Basically, the web server enables users to connect with the main web server which can occur if users are sharing files, which is also done through LAN or WAN, but most commonly a WAN for the majority.

Yet another example of an application service is a web proxy. Basically, a proxy is similar to Virtual Private Network (VPN), which is a way of sending a signal to a computer, which then gets sent to the server. However, a proxy is when the network traffic is intentionally diverted to a completely different server.

Essentially, this allows users to work and stay active as it requires software in order to run successfully in which without the required software, proxy will have problems from running smoothly and quickly.

Overall, Application services is the ability for the user to access files on the network and other software applications. Meaning that this is a broad specification that determines which application help the network carry out one of its main purpose, which is the storage and distribution of data over a network of connected devices.

Shared Resources

Shared resources are resources on a network which the user can have access to. To elaborate, these type of resources can be referred computer data, information or hardware devices. Which can usually be accessed from a device through a local area network or through cabling, such as ethernet cable or other forms of connection cables.

Usually, these resources can be found on computer. This is so that the user can gain access to them, meaning that they would need to be able to connect through the network. The user can share their files to devices such as a printer.

The printer is a type of server which allows the user to print documents as a hard copy, with the ability to control the number of documents which are printed at once. If there are number of user want to print out work, they would need to wait in a queue. So, whoever was the first user who sent their work to the printer for printing, their work would be printed first.

However, printers are one of the few resources which are shared over the network. The most widely used shared resources found in a network come under two categories. Those categories are: Software resources (such as information, files and multimedia) and hardware resources (such as scanners, fax machines, photocopiers and printers).

Meaning that these type of resources can be referred computer data, information or hardware devices. Where these can be accessed from a remote computer which is connected through a local area network or an ethernet/intranet connection.

Therefore, these resources are mostly found on a computer, giving the user the ability to gain access to them with ease. However, to connect to these shared resources would mean to have a stable connection to them, meaning that they would need to be able to connect through the network which they are found or used.

Usually, a shared resource is used by many users in which the most important device that this includes is printing. Shared printing is when a group of workstations are connected to one printer and so, they are able to print out of that one printer.

Therefore, this is extremely useful for a business, due to the fact this means that the business wouldn't have to purchase as many printers to equal that to every workstation. Since a printer can be connected to multiple workstations at once, so this would save the company a lot of money due to the fact a group of computers can connect to a single printer.

For example, using a printer in businesses and other organisations wouldn't be as impactful on the costs. Due to the fact that all or most users (group of workstations) are connected to one printer, giving the ability to print documents using that printer. This will save the cost of investing on a new printer for schools and businesses as they are sharing one printer among different users.

However, this can bring disadvantages of the amount of work is printed out. For instance, if there are number of users who wants to print their work, such as a document. They will have to wait in a queue as the printer will print the document for the first user first. You can also connect the printer from your mobile phone.

Overall, Shared resources are the devices shared between different users on a network, used to reduce cost and for ease of access. Which is done by connecting a single device to multiple workstations within a network, most commonly a LAN.

Basically, shared resources could be anything shared between users. For example, data stored in a server, access to a printer or even information in Google Drive. Making shared resources also a wide collection of devices and software which can be used to share information or accessibility.

Conclusion

Overall, a network is designed to provide services and information over devices connected together via the hardware and software allocated for the structure of a network. These allocated hardware and software tools are the necessary network tools called network services.

Mainly there are six main types of network services. Those services are: Directory services, email, remote access, file sharing, application services and shared resources. These services are vital for the integrity of data storage, integrity, transfer and performance. While also being able to maintain the devices on the network to function at optimal level.

Directory services are the software programs which are connected directly into the core databases, to ensure effective management of the information and security of users found on the network. It is done via software which is designated to keep, organise and private transfer of data in a certain directory. Basically, it allows the lookup of a given name, equal that to a dictionary. Since a single word that is found in a dictionary could have multiple meanings, whereas in a directory, a name would be connected with a multiplex, which is a diverse portion of data.

Email, short for electronic mail, is a telecommunication service which has the ability to send data in the form of text, images, videos and other file attachments in the form of packets to a receiver. This is done by a series of servers that acknowledge, find a transfer location and deliver while still maintaining the integrity of the data in the packets. Hence why an internet connection is vital in order to send an email to another user.

Remote access is another feature of a telecommunication service. It is essentially the ability of an administrator to control a user's device from a distance. This is achieved by successfully activating a remote access point which would give the administrator access to the machine.. However, in order to carry this out the PC needs access to the remote application, such as Remote Access or PC Anywhere. Alos, the user need a password and a username in order to certify the access for the remote access.

File sharing is a private and public distribution of computer information (such as data) in a network, which may contain different standards of access privilege. Also, file sharing isn't a LAN only ability, it could also be used outside a local network. Essentially, it works as a storage space (equal to a hard drive) on a cloud server, so it could be accessed with any device that supports the file shared. However, sharing files is only possible when connected to the internet, but the files can still be accessed and sometimes edited while offline.

Application services are a very broad term for software that gives the ability of managing and storing data, whether it is through software based services or protocols over a network. The reason that it is a very bread term is due to the fact that these services provides communication used in either peer to peer network or server network. Which is the main purpose of a computer network, so the explanation of all the services can be done in a very simplistic manner.

Shared resources, which is also known as network resources, is the sharing of computer information and hardware devices on a computer network. Due to this very broad definition, a lot of software and hardware can fall under this category within a network. Essentially, shared resources is when a single resource (such as a printer) is shared (or connected) to a variety of different workstations (Personal Computers) over a network, giving multiple users the ability to access the single network resource.